Semantic Space Abstractions for Partially Observable Deterministic Task and Motion Planning

Final Presentation

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23rd Jun, 2026

Chair of Machine Learning and Reasoning (Computer Science 6)

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Roadmap

1. Introduction
2. Background
3. Related Work
4. Approach
5. Experimental Evaluation
6. Discussion

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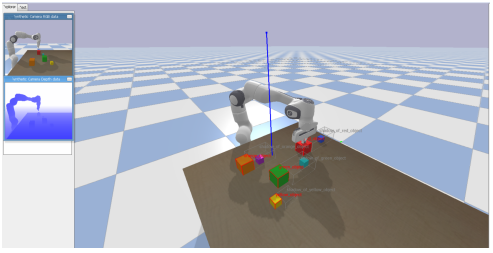
- This needs both multi-step reasoning *and* physical manipulation $\Rightarrow$ **Task and Motion Planning (TAMP)**
- TAMP links a high-level goal to feasible motion: a symbolic task plan, checked against geometry and kinematics
- **But**: classical TAMP (such as PDDLStream) assumes a *fully observable*, *deterministic* world

Introduction: Planning under Occlusion

- Real scenes are **partially observable**: occlusion, limited sensor coverage



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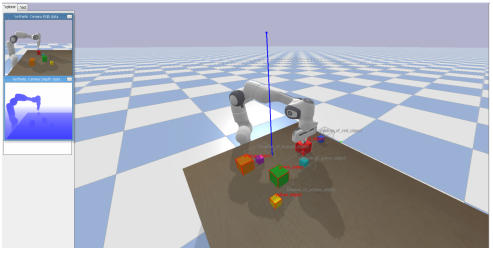

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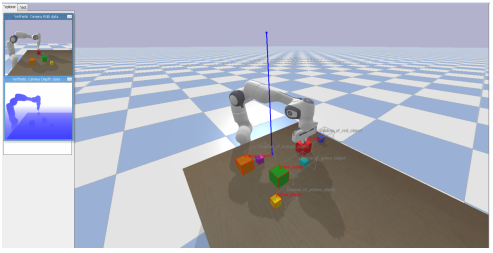

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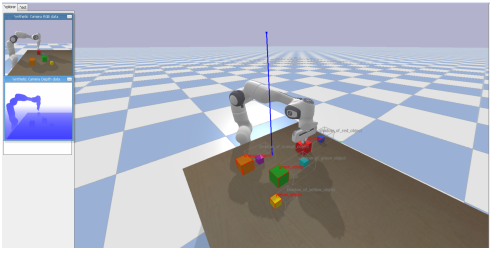

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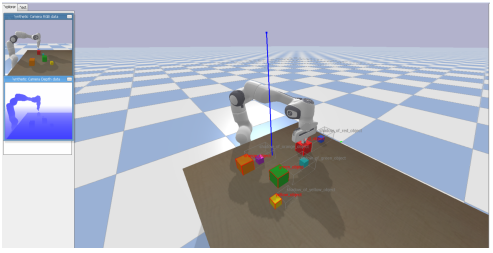

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Research question: How can we define and integrate a belief representation based on *semantic partitioning* into a partially observable deterministic (POD) TAMP framework?

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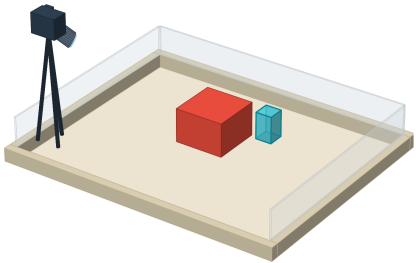

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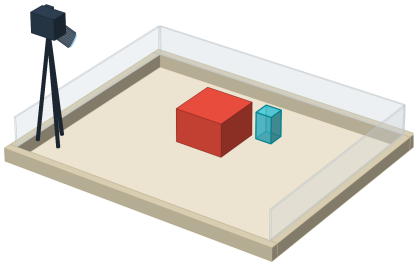

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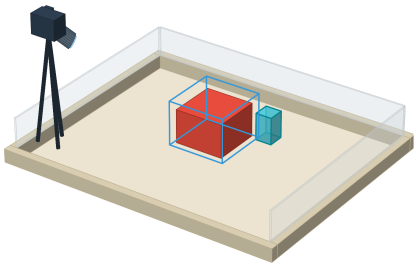

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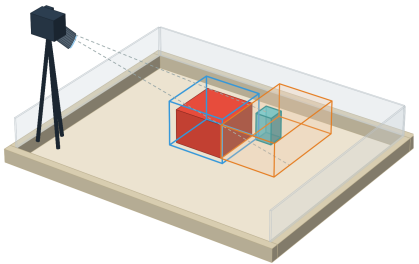

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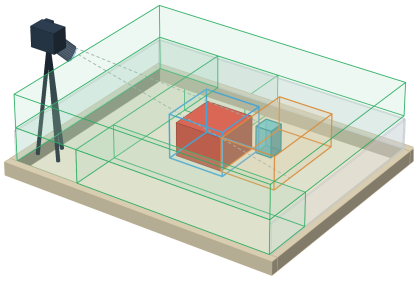

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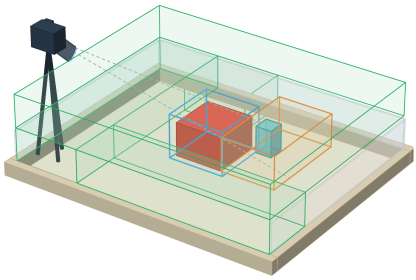

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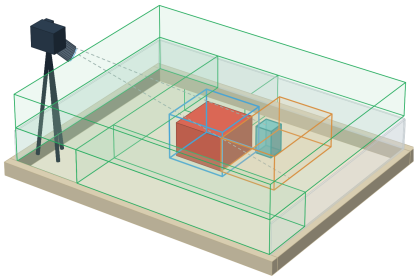
- A classical planner reasons over these few cells, and *looks* only where it must



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- **STRIPS / PDDL** represents it compactly: a *factored* state is the set of true atoms, e.g. $\{$`(on A B)`, `(handempty)`$\}$; an action has preconditions and add / delete effects

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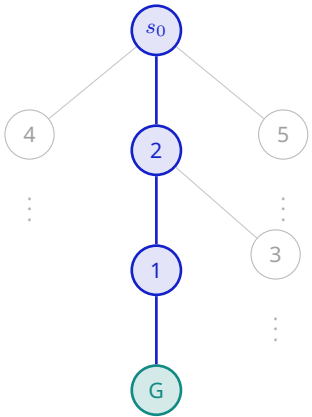
- Expand states **best-first**, ordered by a *heuristic* $h(s)$ — an estimate of the remaining distance to the goal



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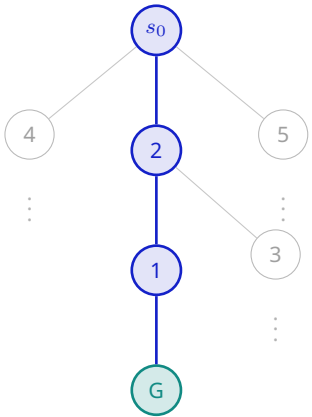
- Promising states go first, so the goal is reached **without enumerating** the whole state space



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Example — a target hidden behind one of three objects:

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Example — "is the cup in region B?"

before sensing B

```
obj_at_boxel(cup,B)
= ?  (don't read it yet)
```

```
obj_at_boxel_KIF(cup,B)
absent
```

value UNKNOWN

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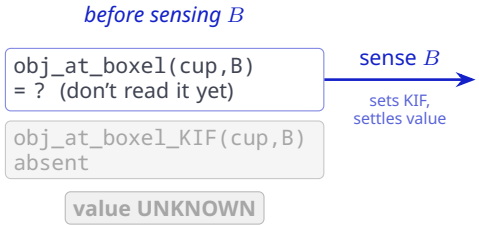

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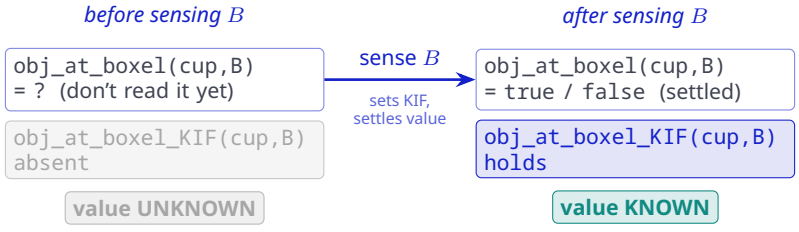

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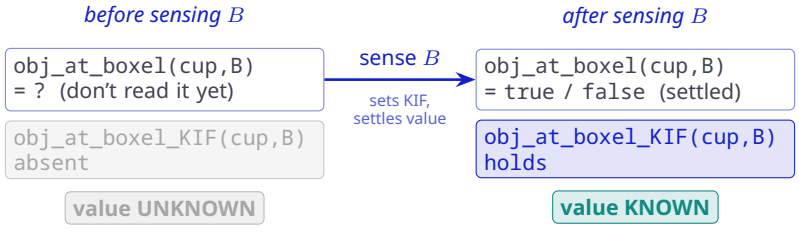

This **compiles** partial observability into a **classical** problem: the planner now seeks a *state of knowledge* — the target's region becomes *known* (Geffner & Bonet [Gef+13]).

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Two very different reasons a plan can be uncertain:

- **Random outcomes** — e.g. a move that occasionally slips sideways. This needs a **POMDP**: a probability distribution over states and a *policy* — expressive, but computationally very demanding (Kaelbling et al. [Kae+98])

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Ours is the second kind: the robot's actions are reliable — the only uncertainty is occlusion. So no probabilities; we track knowledge, and sensing resolves it.

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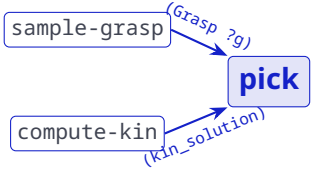
Example — our `pick` action:

```
(:action pick :parameters (?o ?b ?g ?q)
 :precondition (and (handempty)
    (obj_at_boxel ?o ?b)
    (at_config ?q)                    // from move
    (Grasp ?g)                        // sample-grasp
    (kin_solution ?o ?b ?g ?q)))
 :effect (and (holding ?o) ...))
```



Each stream certifies one blue fact `pick` needs. The trajectory stream `plan-motion` feeds the preceding `move`, which sets `at_config`.

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- **Determinize-and-replan**: SS-Replan replans deterministically in belief space on each observation (Garrett et al. [Gar+20]) — we *share* this. The POD **K-replanner** compiles partial observability to classical via knowledge literals (Bonet & Geffner [Bon+11]), but is pure task planning; we bring that line *into TAMP*, on named **Boxel** volumes

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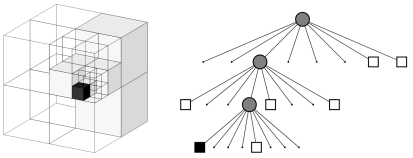
What we look for:

occlusion as *named, symbolic state* a planner can resolve.

- **Octree maps** (OctoMap): map *where matter is*; unobserved space is only absent evidence, and subdivision is geometric, not semantic (Hornung et al. [Hor+13])



An octree map: free (white) and occupied (black) cells, with its tree (after Hornung et al.).

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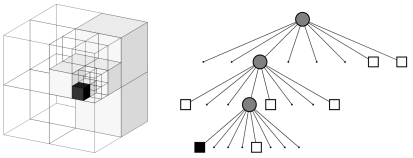

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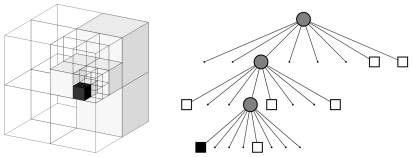
- **Occlusion-aware planners**: keep occlusion in a *probabilistic belief* (visibility grid or pose distribution), not a named volume (Garrett et al. [Gar+20])



An octree map: free (white) and occupied (black) cells, with its tree (after Hornung et al.).

Related Work: Spatial Representations under Occlusion

What we look for:

occlusion as *named, symbolic state* a planner can resolve.

- **Octree maps** (OctoMap): map *where matter is*; unobserved space is only absent evidence, and subdivision is geometric, not semantic (Hornung et al. [Hor+13])
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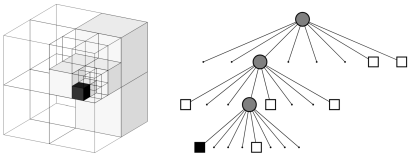


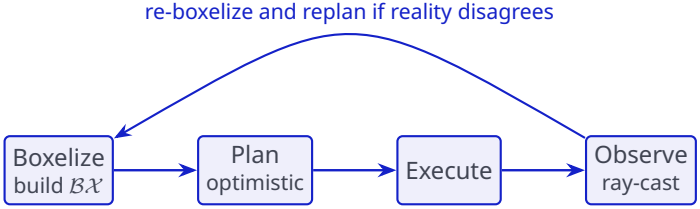
An octree map: free (white) and occupied (black) cells, with its tree (after Hornung et al.).

The gap: none names the occluded *volume* as symbolic state that sensing resolves inside a PDDLStream TAMP loop — what this thesis fills.

Approach: The Sense–Plan–Act Loop

The method, end to end:

boxelize the scene, plan optimistically, execute, observe — then loop. The rest of this section fills in each step.



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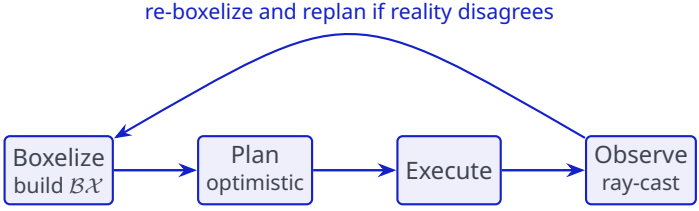

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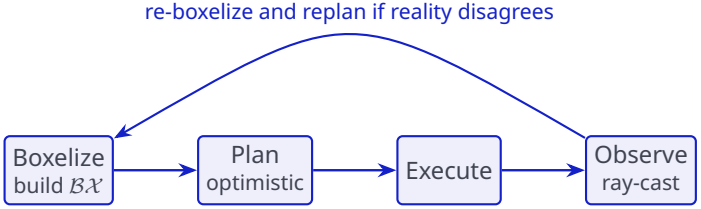

- Plan optimistically, execute, **repair by replanning**
- **Re-boxelize** when a placement consumes a cell or a new occluder appears
- **Time-bounded**: ends on goal, no plan, search exhausted, or budget

Approach: Building Boxels

Start from the objects, the camera, and the workspace — no Boxels yet

1. **Object Boxels**
   wrap each object in a tight box
2. **Shadow Boxels**
   add the region each object hides from the camera
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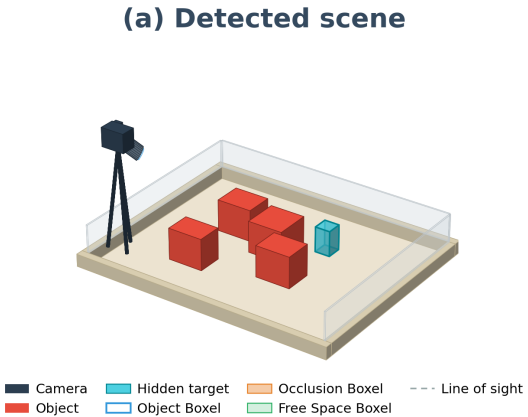

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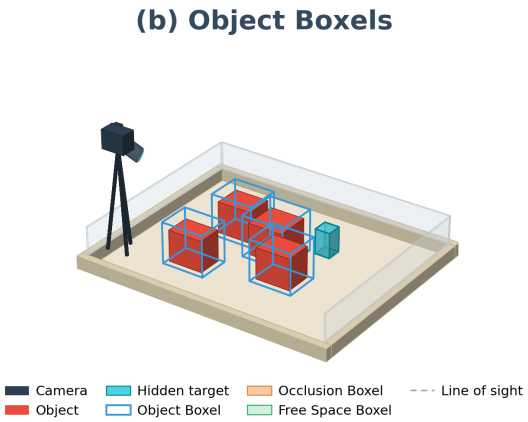

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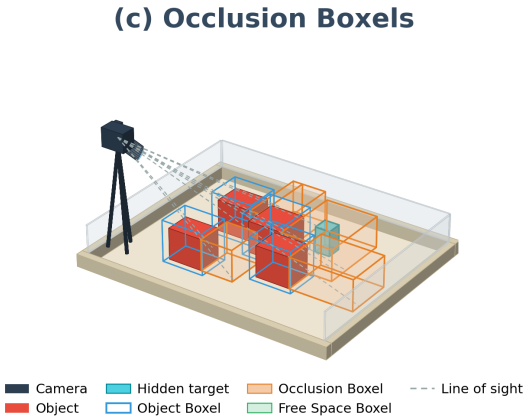

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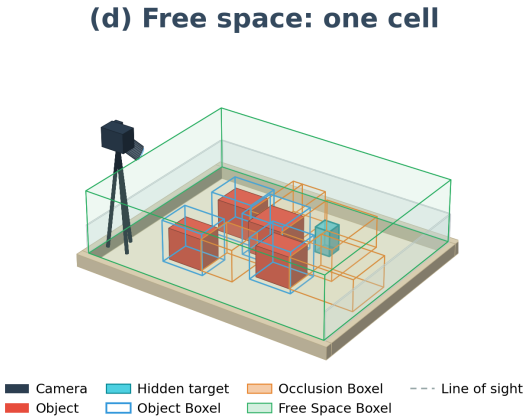

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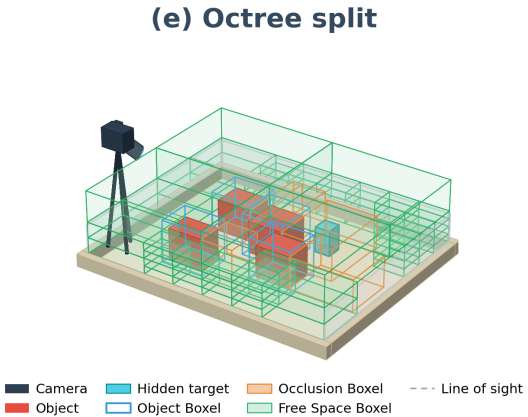

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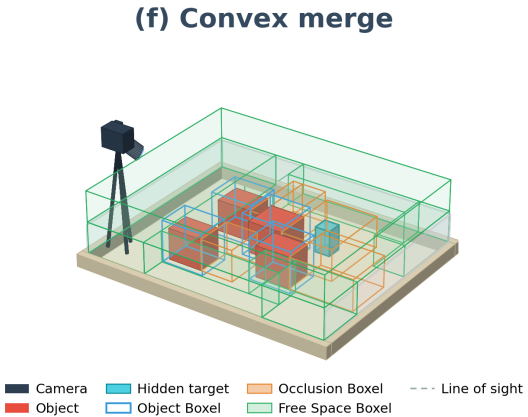

Approach: Belief over Boxels

- The Boxel partition *is* the belief support: a hidden target gives **one candidate world per Boxel** it could occupy ($\mathcal{S}_0$)

Example: target t behind one of 3 occluders

belief $\mathcal{S}_0$: 3 worlds

```
KIF(t,b1) = ?
KIF(t,b2) = ?
KIF(t,b3) = ?
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$\texttt{KIF}(t, b_i)$: is target t in shadow Boxel b_i? The unknown ($?$) ones are the candidate worlds.

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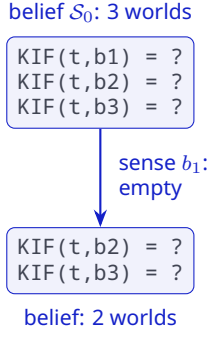

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- Carried by **Know-If fluents**, so a plain classical planner reasons over the hidden object — **no separate belief solver**

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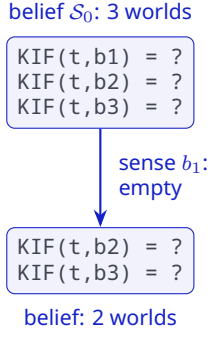

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  - `plan-motion` — a collision-free trajectory (RRT-Connect (Kuffner & LaValle [Kuf+00]))

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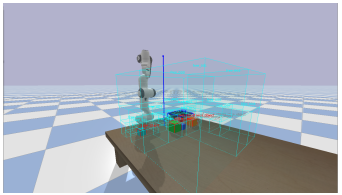

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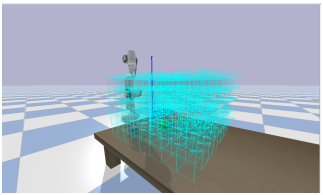

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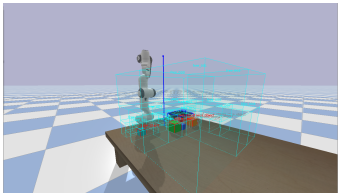

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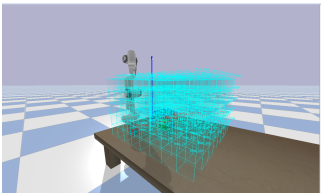

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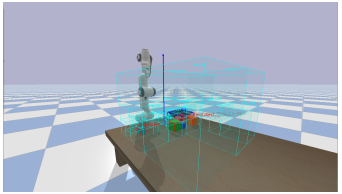

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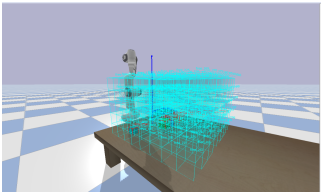

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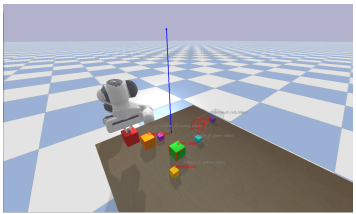
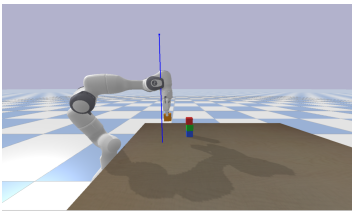
Setup: PyBullet (Coumans & Bai [Cou+21]) + 7-DOF Franka Panda, one fixed camera, **oracle** perception; 3 goals $\times$ 3 difficulty levels $\times$ 30 seeds, 1800 s budget each. Oracle perception isolates the representation — an ablation, not a benchmark win.

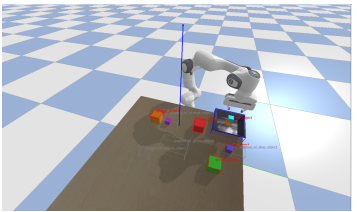
Experimental Evaluation: Tasks



`holding`: retrieve a hidden target



`stack`: build a tower to height h



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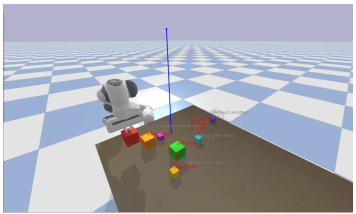

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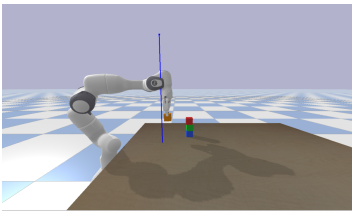

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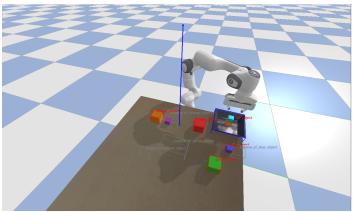
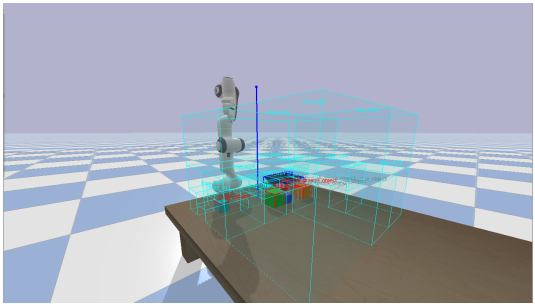


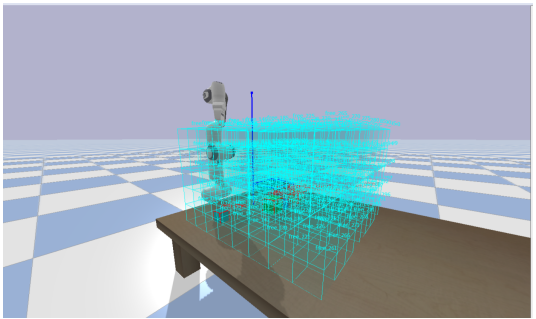
`find-and-tray-stack`: find, then stack on a tray

- Difficulty axis: number of occluders $n_{\text{occ}} \in \{2, 3, 4\}$ (search goals), or stack height $h \in \{2, 3, 4\}$ — each adds shadow regions or tower layers

Results: Semantic vs Uniform Partition



semantic: ≈22–38 task-relevant Boxels



uniform: ∼300 fixed cells over the whole workspace

Same cell size, same planner — the adaptive partition grounds and searches an order of magnitude fewer cells.

Results: Success and Planning Time

goal	variant	success	mean plan time (s)
find-and-tray-stack	**semantic**	**75.6 %**	**124.6**
	uniform	34.4 %	692.1
holding	**semantic**	**65.6 %**	**134.3**
	uniform	47.8 %	438.5
stack	**semantic**	**64.4 %**	**17.2**
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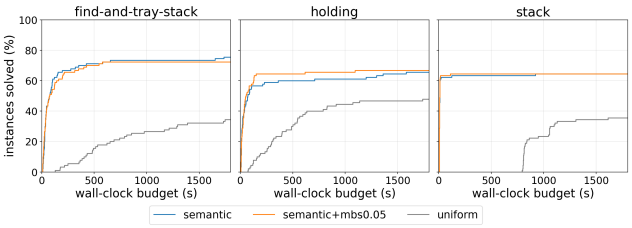
Same planner, same cell size: the adaptive partition solves $\approx$1.4–2.2$\times$ more scenes, while planning several-fold to $\sim$50$\times$ faster.

Results: The Cost Gap



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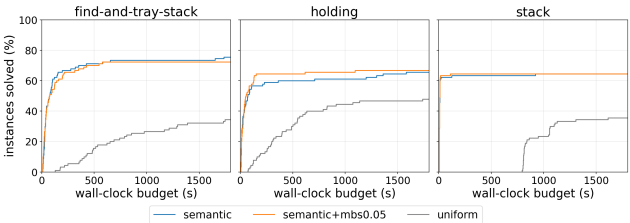

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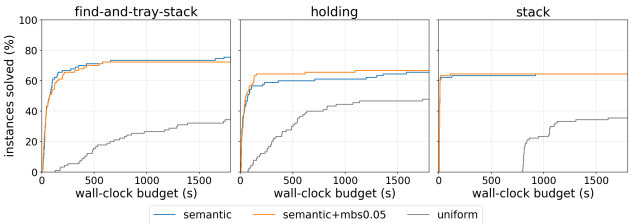
- The adaptive partition **dominates at every budget**; `uniform` starts later and plateaus lower
- **Mechanism:** ≈22–38 Boxels vs ∼300 free cells ⇒ ∼300 vs 11k–14k grounded facts

Results: The Cost Gap



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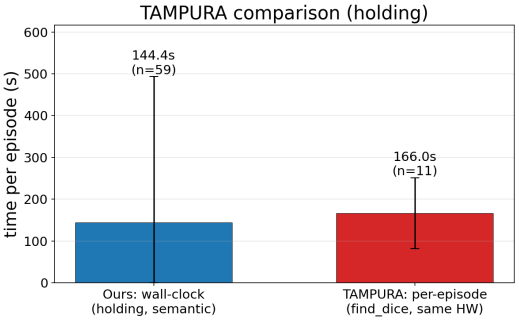
Results: Comparison with TAMPURA



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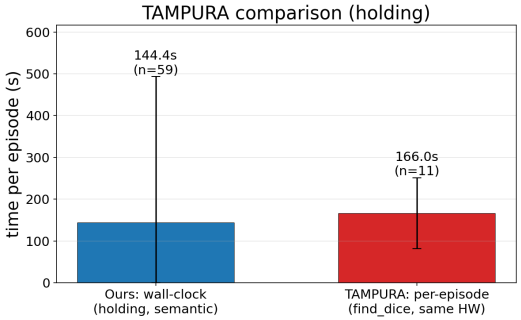

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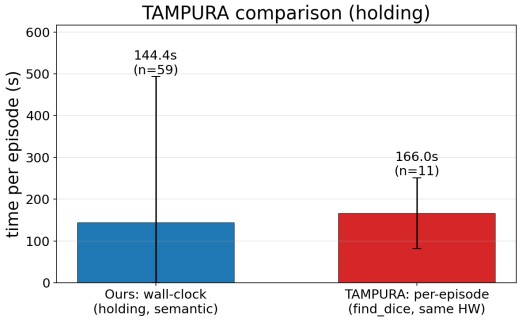

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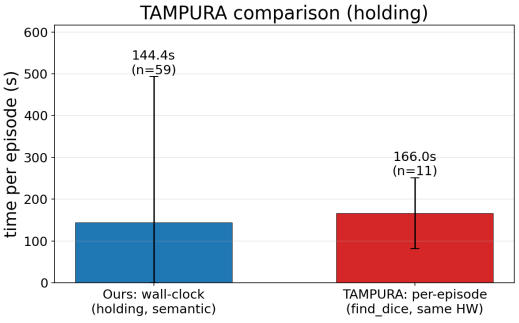

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- We hold occlusion as **symbolic state** — line of sight is a planning condition; TAMPURA keeps it sub-symbolic
- Not a benchmark win: different environments, same hardware

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- **Bounded give-up:** on a permanently blocked shadow the planner eventually stops searching — unsound, since the target may still be reachable (we determinize-and-replan, not full contingent POD)

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  - a full semantic free-space merge

Takeaway: making occlusion an explicit, named piece of symbolic state lets a simple *deterministic* planner do partially observable TAMP — reasoning about where to look — without learning a probabilistic policy.

Summary

- **Problem:** TAMP under occlusion — the robot must decide *where to look*, but a dense voxel belief is too large to plan over
- **Idea: Boxels** — task-relevant box-shaped cells (object / shadow / free) that put the structure of space into the symbolic state
- **Method:** a deterministic POD-TAMP planner reasons over Boxels with Know-If fluents; optimistic sensing + replanning resolve the unknown
- **Result:** an order of magnitude fewer cells $\Rightarrow$ far more cluttered scenes solved than a uniform grid; at parity with TAMPURA, without a learned probabilistic policy

Thank you! Questions?

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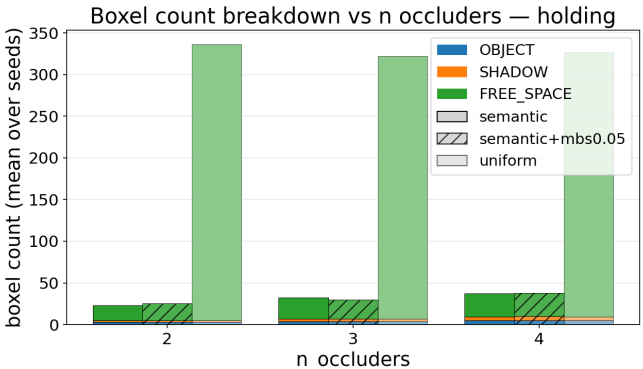
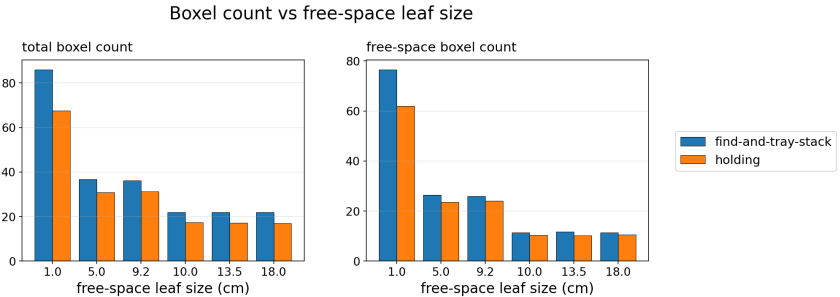
Backup: Representation Compactness



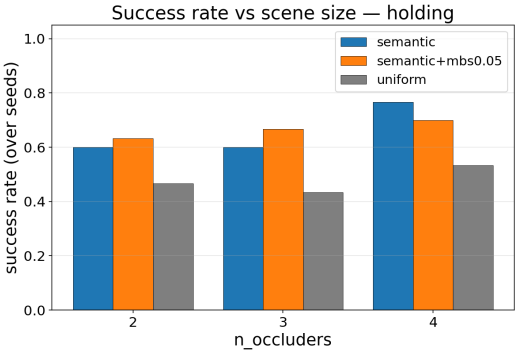
- Adaptive: ≈22–38 Boxels in total
- `uniform`: ∼300 + free-space cells alone
- An order of magnitude fewer cells to ground and search

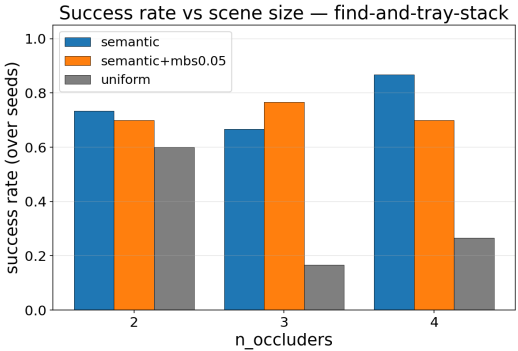
Backup: Free-Space Resolution



The partition is **insensitive** to the free-space leaf size: flat from 5 cm to the auto setting; only the extreme 1 mm floor breaks feasibility (too fine to plan within budget).

Backup: Scaling with Clutter

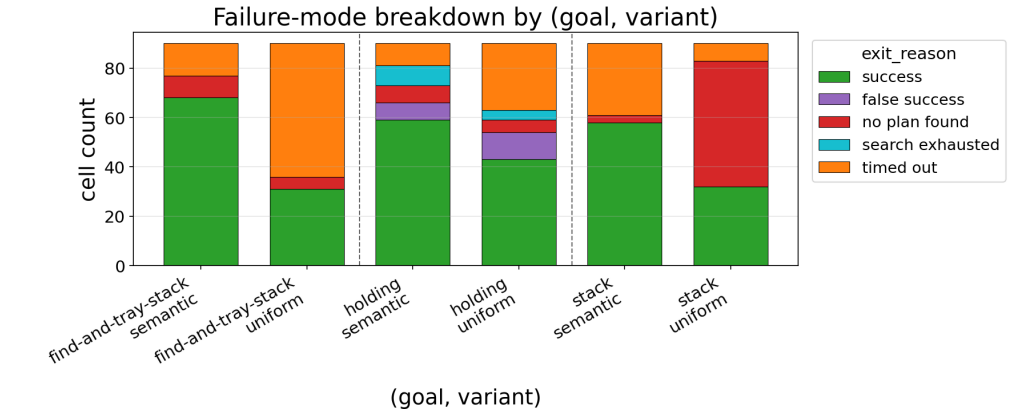




More occluders do not break the adaptive partition: it stays well above `uniform` at every clutter level, and even rises at the densest scenes.

Orange is `semantic+mbs0.05` — the same partition with a 5 cm floor on free-cell size (a resolution check); it tracks `semantic`. `stack` instead scales with tower *height*, not clutter — a manipulation limit, not the planner.

Backup: Failure Modes



Exit reason per (goal, variant), 90 runs each. **uniform** fails mostly by *timing out* (and *no plan found* on `stack`): it cannot ground and search ∼300 cells within budget. **semantic**'s residual failures are fewer, split across *timed out*, *search exhausted* (target outside every box-shaped shadow), and *false success* (a physics mismatch at execution).